



FLORIDA STATE UNIVERSITY
DEPARTMENT OF SCIENTIFIC COMPUTING

WRITTEN PRELIMINARY EXAMINATION
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Question - 10

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1 Numerical Integration[Dr. Sachin]

1.1 Part (a)

The given function to be integrated is

$$f_n(x) = \sum_{j=1}^n \frac{|\sin(j\pi x)|}{j}$$

This function is continuous over the domain $x \in [0, 1]$ but its derivative $f'_n(x)$ is discontinuous.

Points of discontinuity of $f'_n(x)$:

Differentiating $f_n(x)$ with respect to x :

$$f'_n(x) = \frac{d}{dx} \left(\sum_{j=1}^n \frac{|\sin(j\pi x)|}{j} \right)$$

$$f'_n(x) = \sum_{j=1}^n \frac{d}{dx} \left(\frac{|\sin(j\pi x)|}{j} \right)$$

Now, chain rule can be used to calculate the derivative of $|\sin(j\pi x)|$:

$$\begin{aligned} \frac{d}{dx} |\sin(j\pi x)| &= \frac{d}{d(\sin(j\pi x))} |\sin(j\pi x)| \cdot \frac{d}{dx} (\sin(j\pi x)) \\ &= \text{sgn}(\sin(j\pi x)) \cdot (j\pi) \cdot \cos(j\pi x) \end{aligned}$$

where $\text{sgn}(x)$ is the sign function defined as:

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Thus, the derivative of $f_n(x)$ becomes:

$$f'_n(x) = \sum_{j=1}^n \frac{\text{sgn}(\sin(j\pi x)) \cdot (j\pi) \cdot \cos(j\pi x)}{j}$$

Hence,

$$f'_n(x) = \pi \sum_{j=1}^n \text{sgn}(\sin(j\pi x)) \cdot \cos(j\pi x)$$

The derivative $f'_n(x)$ will be discontinuous where the sign of $\sin(j\pi x)$ changes. One such point is $x = 0$ whose discontinuity is straightforward to show (I have done that at the end in **Appendix**). Other points of discontinuity occur when x crosses multiples of $\frac{1}{j}$, for integer values of j . So, the points of discontinuity of $f'_n(x)$ are at the points:

$$x = \frac{k}{j}$$

where k is an integer and j ranges from 1 to n .

For $n = 30$, the number of such discontinuities can be calculated as follow:

For each j from 1 to n , we need to count how many valid k there are such that $0 \leq \frac{k}{j} \leq 1$:

- For $j = 1$, k can be 0 or 1. This gives 2 points.
- For $j = 2$, k can be 0, 1, 2. This gives 3 points.
- For $j = 3$, k can be 0, 1, 2, 3. This gives 4 points.
- ...
- For $j = n$, k can be 0, 1, 2, ..., n . This gives $n + 1$ points.

In general, for a given j , the number of discontinuity points is $j + 1$.

The total number of points is the sum of all discontinuity points from $j = 1$ to n :

$$\sum_{j=1}^n (j + 1) = \sum_{j=1}^n j + \sum_{j=1}^n 1$$

This is easy to calculate and the summation is equal to 495. So, there are 495 points of discontinuity of $f'_n(x)$. The same is verified using the Python code in 10.ipynb.

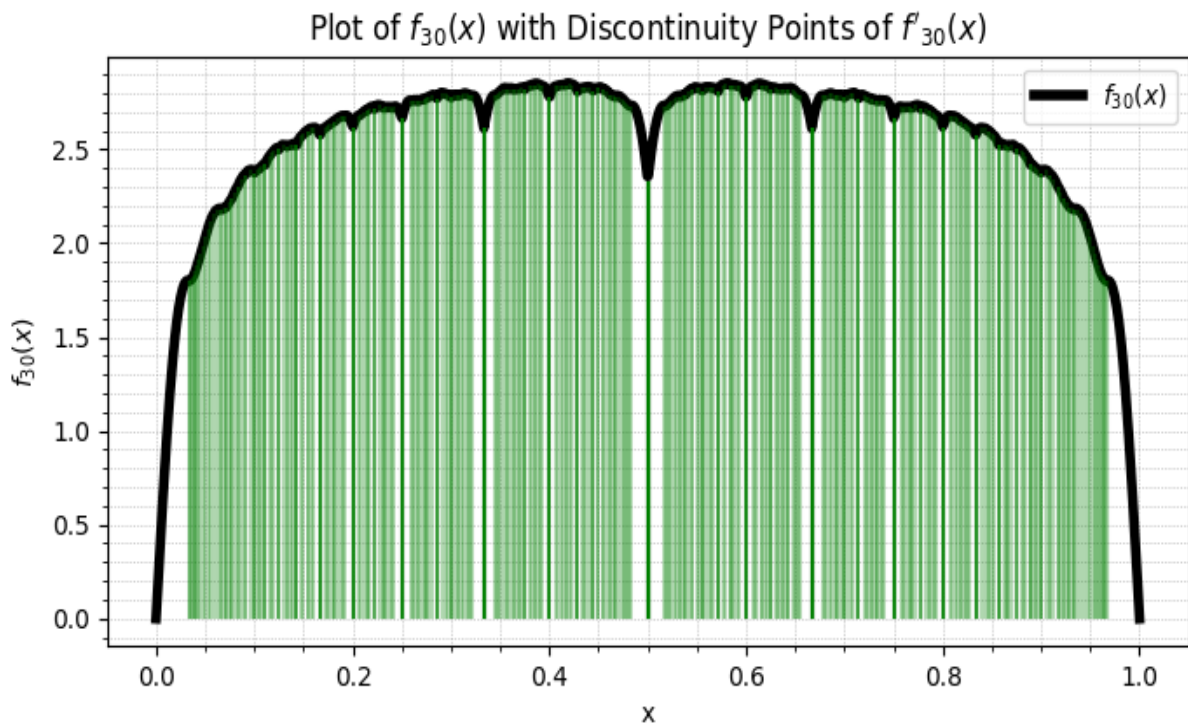


Figure 1: Plot of function $f_{30}(x)$ with vertical lines at discontinuity points.

1.2 Part (b)

I have implemented two methods: Trapezoidal integration and method of adaptive quadrature using `quad` from `scipy`. I would prefer the `quad` function for this integration task due to its high accuracy and adaptive nature, which effectively handles the discontinuities in the derivative of the integrand.

The analytical results (**shown together at the end**) confirms the accuracy of the numerical estimate obtained using `quad`. Other adaptive quadrature method such as Gauss Legendre quadrature can also be effective as they do not divide the whole domain into equally spaced points. Adaptive quadrature methods do importance sampling.

I have considered following points [1] for making my choice:

- The **quad** function uses an adaptive algorithm that dynamically subdivides the integration interval which is crucial for handling regions with sharp changes or discontinuities. And as shown in fig-1, there are lots of irregularities and change in values of the given function. So, this adaptive nature is quite beneficial.
- The adaptive algorithm increases the density of points in problematic regions, effectively dealing with discontinuities in the derivative.

Results:

- **Trapezoidal Rule:** ≈ 2.5309
- **quad (Adaptive Quadrature):** ≈ 2.5434
- **Gaussian Quadrature:** ≈ 2.5422
- **Analytical Result:** ≈ 2.5433

1.3 Part (c)

The integral to be calculated analytically is

$$I = \int_0^1 f_{30}(x) dx,$$

where function f_{30} is a summation series:

$$f_{30}(x) = \sum_{j=1}^{30} \frac{|\sin(j\pi x)|}{j}.$$

We can write the integral as follow now:

$$I = \int_0^1 \sum_{j=1}^{30} \frac{|\sin(j\pi x)|}{j} dx = \sum_{j=1}^{30} \int_0^1 \frac{|\sin(j\pi x)|}{j} dx.$$

Each term in the summation can be integrated separately:

$$\int_0^1 \frac{|\sin(j\pi x)|}{j} dx.$$

To integrate $\int_0^1 \frac{|\sin(j\pi x)|}{j} dx$, let us first consider the following integral:

$$\int_0^1 |\sin(j\pi x)| dx.$$

For $j \geq 1$, the sine function completes half a period over the interval $[0, \frac{1}{j}]$ and then repeats itself. Thus, $|\sin(j\pi x)|$ is symmetric and periodic.

$$\implies \int_0^1 |\sin(j\pi x)| dx = 2 \int_0^{\frac{1}{2j}} \sin(j\pi x) dx.$$

Now, consider the case of $|\sin(4\pi x)|$. As shown in fig-2, the function $|\sin(4\pi x)|$ is symmetric over $[0, \frac{1}{2 \times 4}]$. From the figure, following information can also be concluded by observing the symmetry:

$$\int_0^{\frac{1}{2}} |\sin(4\pi x)| dx = 4 \int_0^{\frac{1}{2 \times 4}} \sin(4\pi x) dx.$$

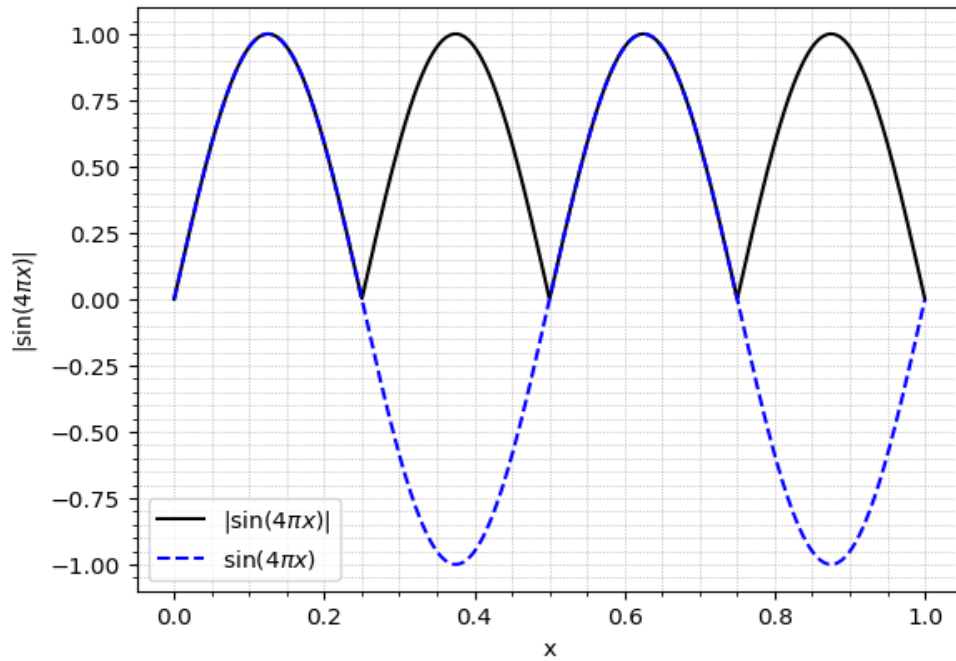


Figure 2: Plot to demonstrate symmetry of $|\sin(4\pi x)|$

This observation can be extended for general case of j and we can write following:

$$\int_0^{\frac{1}{2}} |\sin(j\pi x)| dx = j \int_0^{\frac{1}{2j}} \sin(j\pi x) dx = j \times \left(\frac{1}{j\pi}\right) = \frac{1}{\pi}$$

$$\Rightarrow \int_0^1 |\sin(j\pi x)| dx = \frac{2}{\pi}.$$

So, each integral becomes:

$$\int_0^1 \frac{|\sin(j\pi x)|}{j} dx = \frac{2}{\pi j}.$$

Thus, the total integral I is:

$$I = \sum_{j=1}^{30} \int_0^1 \frac{|\sin(j\pi x)|}{j} dx = \sum_{j=1}^{30} \frac{2}{\pi j} = \frac{2}{\pi} \sum_{j=1}^{30} \frac{1}{j}.$$

Now, using Wolfram Alpha (also done using Python in my notebook),

$$\sum_{j=1}^{30} \frac{1}{j} \approx 3.9951$$

Thus, the integral I is approximately

$$\Rightarrow I \approx \frac{2}{\pi} \times 3.9951.$$

$$I \approx 2.5422.$$

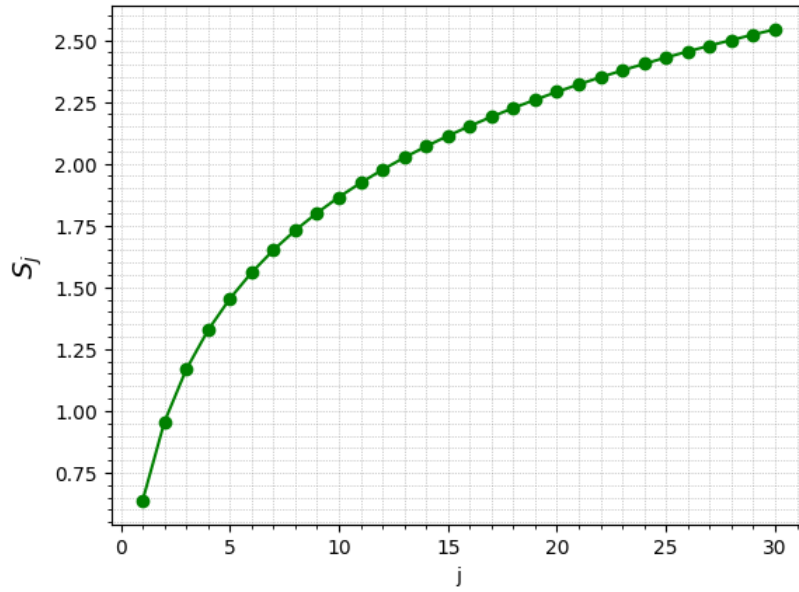


Figure 3: Integration result as each term is added after integrating.

The numerical result from the `quad` function (2.5434) closely matches the analytical result (2.5433), indicating that the `quad` function provides a very accurate approximation. The trapezoidal rule result (2.5309) is slightly less accurate but still reasonably close.

1.4 Appendix: Discontinuity of $f'_n(x)$ at $x = 0$

As x approaches 0 from the left,

$$\begin{aligned}
 \lim_{x \rightarrow 0^-} f'_n(x) &= \lim_{x \rightarrow 0-h} f'_n(x) \\
 &= \lim_{h \rightarrow 0} \sum_{j=1}^n \frac{\cos(-j\pi h) \cdot \pi \cdot \operatorname{sgn}(\sin(-j\pi h))}{j} \\
 &= \lim_{h \rightarrow 0} \sum_{j=1}^n \frac{\cos(j\pi h) \cdot \pi \cdot \operatorname{sgn}(-\sin(j\pi h))}{j} \\
 &= - \sum_{j=1}^n \frac{\pi}{j}
 \end{aligned}$$

As x approaches 0 from the right,

$$\begin{aligned}
 \lim_{x \rightarrow 0^+} f'_n(x) &= \lim_{x \rightarrow 0+h} f'_n(x) \\
 &= \lim_{h \rightarrow 0} \sum_{j=1}^n \frac{\cos(j\pi h) \cdot \pi \cdot \operatorname{sgn}(\sin(j\pi h))}{j} \\
 &= \lim_{h \rightarrow 0} \sum_{j=1}^n \frac{\cos(j\pi h) \cdot \pi \cdot \operatorname{sgn}(\sin(j\pi h))}{j} \\
 &= \sum_{j=1}^n \frac{\pi}{j}
 \end{aligned}$$

Since the left-hand limit and the right-hand limit of $f'_n(x)$ are not equal, the derivative is discontinuous at $x = 0$.

References

- [1] Víctor Elvira, Luca Martino, and Pau Closas. Importance gaussian quadrature. *IEEE Transactions on Signal Processing*, 69:474–488, 2020.
- [2] <https://www.derivative-calculator.net/>
- [3] https://en.wikipedia.org/wiki/Trapezoidal_rule
- [4] ACSI Lecture Notes, Dr. Huang